

A Study on the Fitness of GA for Improving SP Decoding Performance

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We have studied low-density parity-check (LDPC) coding and iterative decoding methods for shingled magnetic recording (SMR) in high-capacity hard disk drives (HDDs), which is an areal recording density of 4 Tbits/inch². Previously, we applied a neural network to evaluate the log-likelihood ratio (LLR) associated with row operations in the sum-product (SP) decoder for LDPC codes. Then, we emphasized the LLR considering the influence of noise depending on the recording pattern by providing the LLRs for the decoding target and its adjacent bits to the neural network in SP decoding. Furthermore, we explored the optimal parameters to emphasize the LLR by applying the genetic algorithm (GA). In this study, to explore more optimal parameters, we propose a fitness function to improve selection accuracy and increase the number of eligible LLRs. Then, we aim to improve the performance of SP decoding by the effect of GA. As a result, by applying the proposed fitness function to the GA, it is possible to maintain high selection accuracy and to increase the probability that LLRs are emphasized in SP decoding. Therefore, it achieves error-free performance with fewer iterations of turbo equalization compared to the case of a conventional fitness function.

Index Terms—Genetic algorithm (GA), low-density parity-check (LDPC) code, neural network, shingled magnetic recording (SMR), sum-product (SP) decoding.

I. INTRODUCTION

IN RECENT years, the explosive increase in data volume has required even higher capacity hard disk drives (HDDs). Therefore, we are focusing on the shingled magnetic recording (SMR) [1], which enhances perpendicular magnetic recording (PMR), and developing signal processing methods to achieve higher recording density. We studied the performance improvement of low-density parity-check (LDPC) coding and iterative decoding methods in SMR [2]. In the SMR, due to its narrow tracks, the signal processing methods are required to reduce the influence of intertrack interference (ITI) and signal-dependent noise, such as transition jitter. We have achieved a reduction of the effects of ITI by applying a 2-D finite impulse response (TD-FIR) filter [3]. Also, to consider the influence of signal-dependent noise, we proposed the sum-product (SP) decoder, in which a neural network evaluates the log-likelihood ratio (LLR) related to row operations using the LLRs of the decoding target bit and its adjacent bits and emphasizes the LLR based on the neural network outputs [4]. Furthermore, we demonstrated that a genetic algorithm (GA) with a fitness function to minimize a bit error rate (BER) in the SP decoder is useful for determining parameters, such as thresholds and weights to emphasize LLRs. However, we believe that setting a fitness function so that more reliable LLRs are emphasized will contribute to improve decoding performance.

In this study, to explore more optimal emphasis parameters, we propose a fitness function to improve the selection accuracy and increase the probability that LLRs are emphasized. Then,

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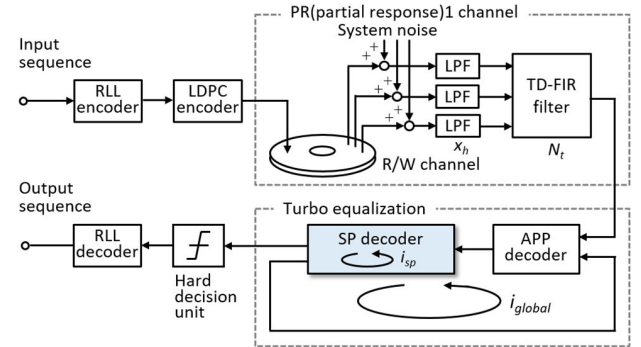


Fig. 1. Block diagram of SMR R/W system with the LDPC coding and iterative decoding.

we aim to improve the performance of SP decoding by the effect of GA.

II. READ/WRITE SYSTEM

Fig. 1 shows a block diagram of an SMR read/write (R/W) system with LDPC coding and iterative decoding. The input sequence is encoded by a 128/130(0, 16/8) run-length limited (RLL) [5] encoder and a (30, 3)-regular LDPC encoder [2]. After that, the recording sequence is recorded on a granular medium model consisting of the discrete Voronoi diagram [6]. In this study, we assume the areal recording density of 4 Tbits/inch². The model sets the average grain size, grain diameter dispersion, and average nonmagnetic grain boundary to 5.0, 1.0, and 1.0 nm, respectively. The system noise defined by the signal-to-noise ratio (SNR) is added at the reading point defined by the following equation:

$$\text{SNR}_S = 20 \log_{10} (A/\sigma_s) \text{ [dB]} \quad (1)$$

where A and σ_s are the positive saturated level of the reproducing waveform from an isolated magnetic transition

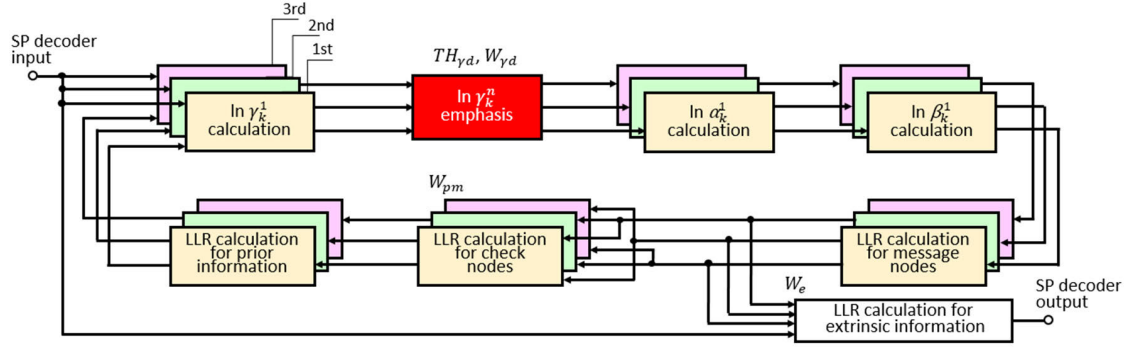


Fig. 2. Block diagram of the SP decoder shown in Fig. 1. The LDPC code of the parity check matrix with a code length of 36 990 bits, a column weight of 3, and a row weight of 30 is employed in this study. As the parity check matrix consists of three submatrices, the LLRs are calculated for the three submatrices.

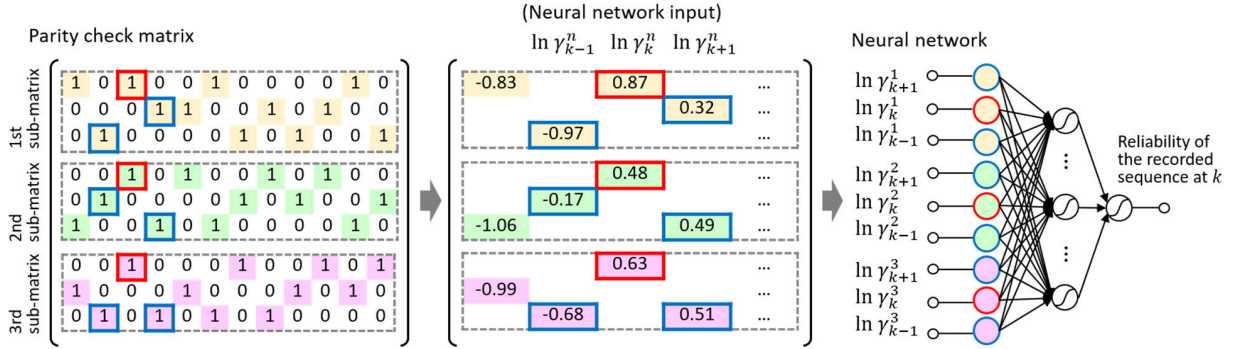


Fig. 3. Relationship between the parity check matrix and the neural network configuration. Here, for simplicity, a case is illustrated in which the code length, the column weight, and the row weight are 12, 3, and 4, respectively.

and the root-mean-square value of the system noise in the bandwidth of the channel bit rate f_c , respectively. The system shows that information from three tracks, the target track and adjacent tracks, is read. We assume the reader sensitivity of a full-shielded reader calculated by the TD-finite element method [3], [7]. A channel bit response of the R/W channel is equalized to the partial response class-I (PR1) target by the equalizer composed of three low-pass filters (LPFs) having cutoff frequency x_h normalized by f_c and the TD-FIR filter with N_t taps, where N_t is the number of taps [3] for each reader. We assume that these parameters x_h and N_t are set to 0.4 and 15, respectively. The turbo equalization is performed between an a posteriori probability (APP) decoder and the SP decoder using a neural network. Additionally, the parameters i_{sp} and i_{global} stand for the maximum number of iterations in the SP decoder and turbo equalization, respectively. After all iterations, the output sequence is obtained through a hard decision unit and the RLL decoder. Then, BER is calculated by comparing the input sequence and output sequence.

III. ITERATIVE DECODING IN THE SP DECODER

In this section, we explain the details of the iterative decoding in the SP decoder proposed in this article.

A. SP Decoding

Fig. 2 illustrates the block diagram of the SP decoder shown in Fig. 1. The LDPC code of the parity check matrix with a code length of 36 990 bits, a column weight of 3, and a row weight of 30 is employed in this study. As the parity

check matrix is composed of three submatrices, the LLRs are calculated for the three submatrices. The light-yellow, light-green, and pink blocks in the figure mean the LLR calculation for the 1st, 2nd, and 3rd submatrices, respectively. Furthermore, the extrinsic information as prior information for APP decoding is calculated from the LLRs for message nodes and SP decoder input and is returned to the APP decoder.

The LLR for the message node is calculated by the Max-log-MAP algorithm [8] using $\ln \gamma_k^n$, $\ln \alpha_k^n$, and $\ln \beta_k^n$ ($n = 1-3$) based on the LDPC code constraints, where γ_k^n means the transition probability from the previous point to the current point on the trellis diagram for the n th submatrix of the parity check matrix. Also, α_k^n and β_k^n mean the forward and backward probabilities on the trellis diagram for the n th submatrix, respectively. Using a method similar to [4], the neural network evaluates $\ln \gamma_k^n$ related to the decoding target bit and its adjacent bits in row operations on the parity check matrix to consider the influence of noise depending on the recording patterns. Then, the reliable LLRs are emphasized using the parameters explored by the GA, as shown in Section III-B.

During the iterative decoding in the SP decoder, the LLR for the check node of each submatrix is obtained by adding the LLRs for the message node of other submatrices, and a priori information for each submatrix is obtained by multiplying W_{pm} based on the parity check information m ($= 0-3$). We discuss the details of the parity check information and the LLR calculation for prior information in [9], respectively. Also, when the parity checks for all rows of the matrices pass or the number of iterations in the SP decoder achieved i_{sp} , the LLR for the extrinsic information is obtained by multiplying

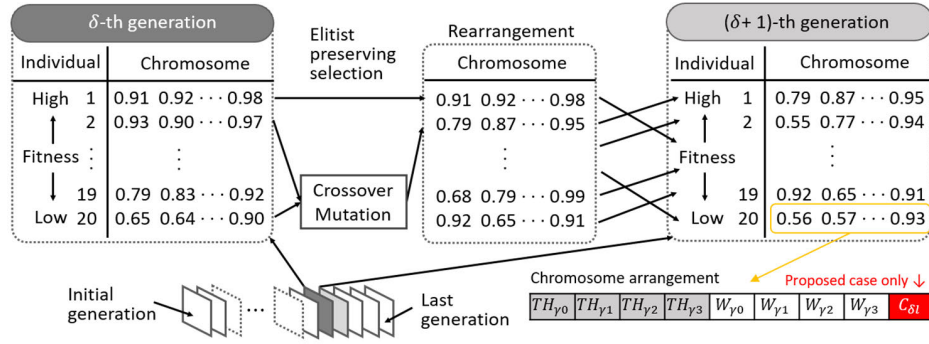


Fig. 4. Concept of GA for exploring the emphasis parameters. We assume a 1-D array of the weights as one chromosome and perform the genetic process.

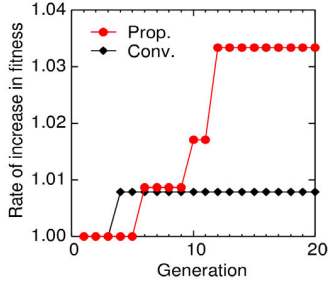


Fig. 5. Rate of increase in fitness.

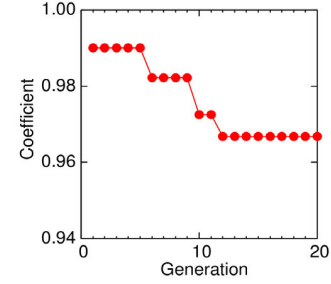


Fig. 6. Coefficient that determines the degree to which the number of emphasized LLRs is reflected.

W_e with the difference between the sum of the LLRs for message nodes and the SP decoder input. Note that we use W_{pm} and W_e explored by the GA, as shown in [9].

B. Exploration of the Emphasis Parameter by the GA

Fig. 3 shows the relationship between the parity check matrix and the neural network configuration. Here, for simplicity, a case is illustrated in which the code length, column weight, and row weight are 12, 3, and 4, respectively. Therefore, there are three LLRs in the column direction and four LLRs in the row direction, respectively. However, we show a part of LLRs in Fig. 3. The LLR surrounded by the red line corresponds to $\ln \gamma_k^n$, and the LLR surrounded by the blue line corresponds to $\ln \gamma_{k-1}^n$ or $\ln \gamma_{k+1}^n$. The neural network calculates the reliability of the recorded sequence at k using $\ln \gamma_{k-1}^n$, $\ln \gamma_k^n$, and $\ln \gamma_{k+1}^n$. The $\ln \gamma_k^n$ emphasis part is determined by whether the neural network output is greater than $TH_{\gamma d}$ or smaller than $(1 - TH_{\gamma d})$. Also, $\ln \gamma_k^n$ for $n = 1-3$ is emphasized by multiplying $W_{\gamma d}$ ($d = 0-3$), respectively [4]. Here, d is the decoding pattern number. The decoding patterns in the cases of $d = 0-3$ correspond to “000” or “111,” “001” or “110,” “011” or “100,” and “010” or “101,” respectively. Note that the parameters, $TH_{\gamma d}$ and $W_{\gamma d}$, are those for emphasis and not those in the neural network.

Fig. 4 shows the concept of GA for exploring the emphasis parameters. In this study, the proposed fitness of the l th individual on the δ th generation is defined by the following equation:

$$f_{\delta l}^{(1)} = -\log_{10}(N_{te}^{\delta l}/N_t^{\delta l}) - C_{\delta l} \log_{10}(1 - N_t^{\delta l}/N_{all}^{\delta l}) \quad (2)$$

where $N_t^{\delta l}$ and $N_{te}^{\delta l}$ are the number of emphasized and incorrectly emphasized LLRs, respectively. $N_{all}^{\delta l}$ is the total number of LLRs in row operations. $C_{\delta l}$ is the coefficient that

determines the degree to which the number of emphasized LLRs is reflected. On the other hand, the conventional fitness as shown in [4] is defined by the following equation:

$$f_{\delta l}^{(2)} = -\log_{10} \text{BER}_{sp} \quad (3)$$

where BER_{sp} is calculated by comparing the LDPC encoder input sequence and hard decision output sequence. As can be seen from Fig. 4, we arrange $TH_{\gamma d}$ and $W_{\gamma d}$ in one dimension as one chromosome. Here, in applying the proposed fitness function, we also include $C_{\delta l}$. The GA performs the selection, crossover, and mutation [4]. However, the individuals with the highest fitness are left by applying elitist preserving selection. In this study, for performing the GA, both the iteration parameters of i_{sp} and i_{global} are set to 20, and SNR_S is set to 21.0 dB. Also, the initial parameters, $TH_{\gamma d}$ and $W_{\gamma d}$, are set to 1.30 and 0.90, based on the parameters considered in [4] when GA was not applied. Then, $C_{\delta l}$ is initially set to 0.99, so that it is almost the same as the initial BER_{sp} in the case of the conventional fitness function. Here, BER_{sp} depends not only on the performance of the SP decoder but also on the performance of the PR channel, as shown in Fig. 1. Therefore, if we use (3), there is a concern that a direct effect may not be sufficient in exploring for emphasis parameters to improve the performance of the SP decoder. On the other hand, if we use (1), which is defined to satisfy both demands to improve the selection accuracy and increase the probability that LLRs are emphasized, we can expect a direct effect on the exploration of emphasis parameters to improve the performance of the SP decoder. We have previously shown that emphasizing more reliable LLRs in SP decoding contributes to effective iterative decoding [4]. Therefore, in this study, we believe it is justified to design the fitness function of

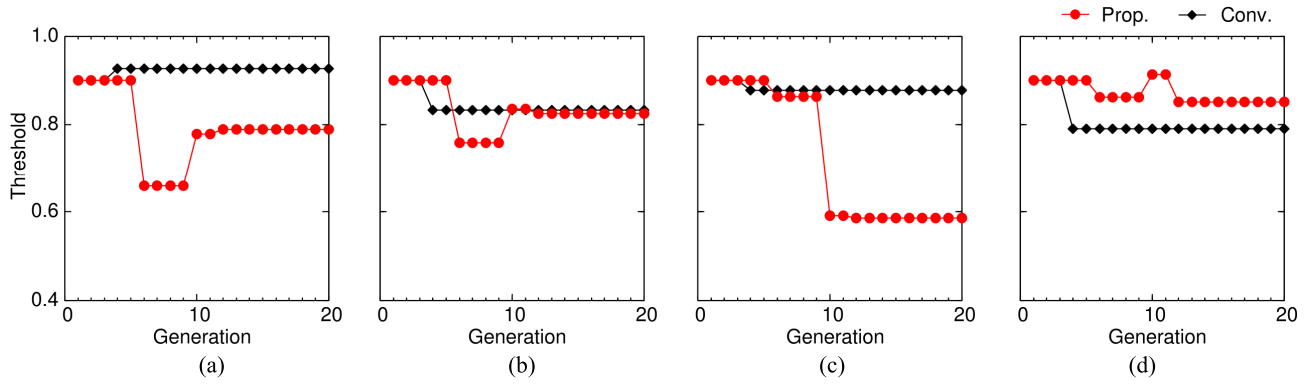


Fig. 7. Threshold, which is the emphasis parameter, based on the decoding patterns. (a) Decoding pattern “000” or “111.” (b) Decoding pattern “001” or “110.” (c) Decoding pattern “011” or “100.” (d) Decoding pattern “010” or “101.”

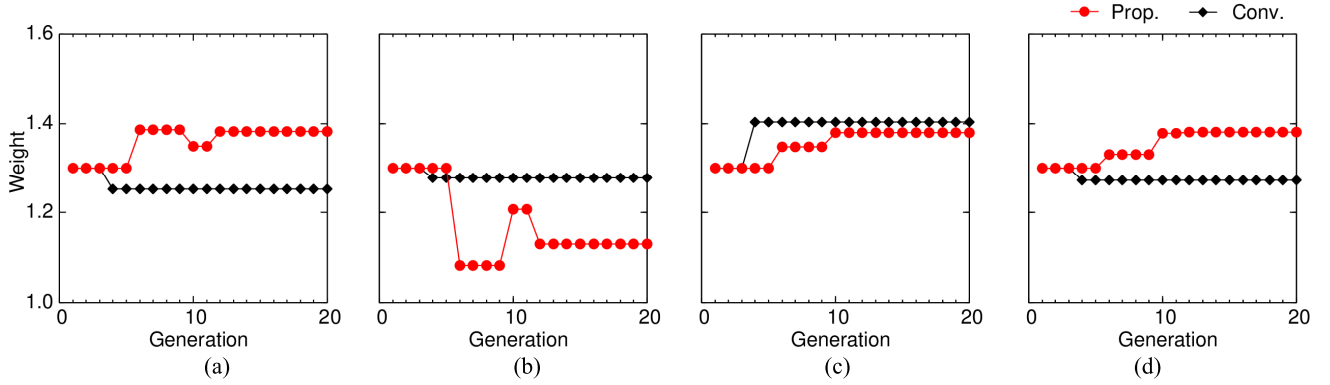


Fig. 8. Weight, which is the emphasis parameter, based on the decoding patterns. (a) Decoding pattern “000” or “111.” (b) Decoding pattern “001” or “110.” (c) Decoding pattern “011” or “100.” (d) Decoding pattern “010” or “101.”

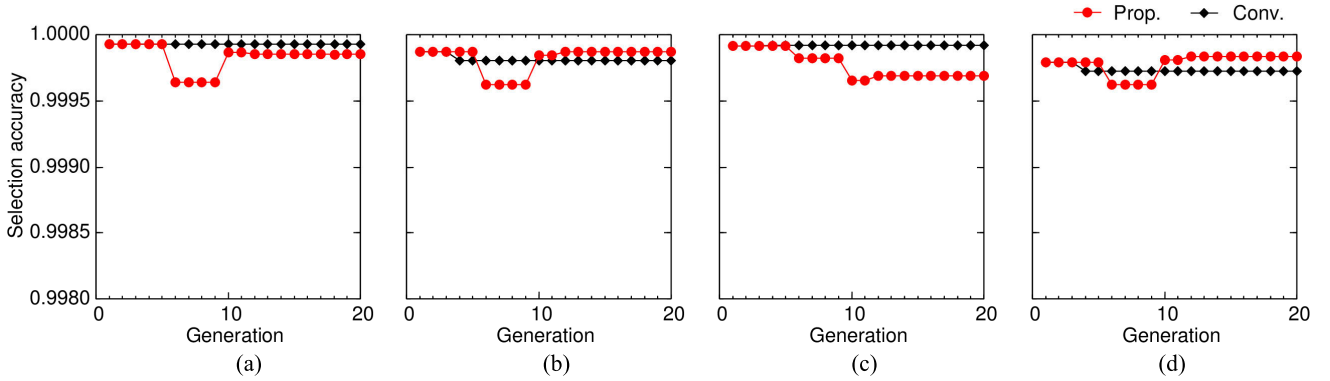


Fig. 9. Selection accuracy of emphasized LLRs based on the decoding patterns. (a) Decoding pattern “000” or “111.” (b) Decoding pattern “001” or “110.” (c) Decoding pattern “011” or “100.” (d) Decoding pattern “010” or “101.”

the GA to improve the selection accuracy and increase the probability that LLRs are emphasized.

IV. PERFORMANCE EVALUATION

First, we evaluate the performance of the best individuals obtained by the proposed or the conventional fitness function. Fig. 5 shows the rate of increase in fitness relative to the first generation. In Fig. 5, the filled-red circle and filled-black diamond marks indicate the cases of the proposed and conventional fitness, respectively. From now on, we will use the same marks. In this study, we consider that fitness converged after about ten consecutive generations and discuss results for 20 generations. From Fig. 5, the rate of increase in the proposed fitness is greater than the conventional fitness

after the 6th generation. Fig. 6 shows the coefficient that determines the degree to which the number of emphasized LLRs is reflected. From Fig. 6, we believe that to balance the selection accuracy and the probability that LLRs are emphasized, C_{sl} converges to 0.965 after 11th generations.

Next, we discuss the results for each decoded pattern, which corresponds to “000” or “111,” “001” or “110,” “011” or “100,” and “010” or “101,” using (a), (b), (c), and (d) of Figs. 7–10. Figs. 7 and 8 show the threshold and the weight as the emphasis parameters, based on the decoding pattern, respectively. From Fig. 7, the proposed fitness function gives a lower threshold, except for “010” and “101,” compared to the conventional one, which contributes to increasing the probability that LLRs are emphasized. From Fig. 8, the

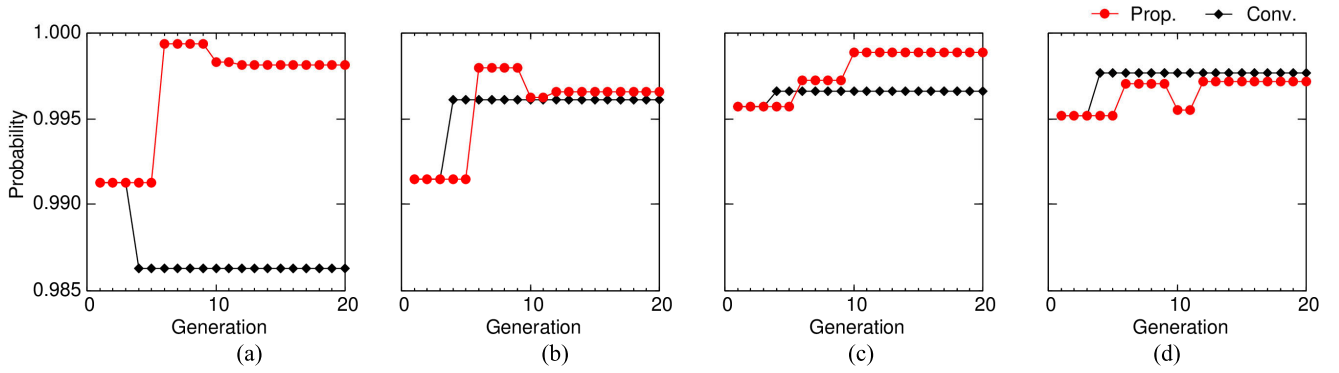


Fig. 10. Probability that LLRs are emphasized based on the decoding patterns. (a) Decoding pattern “000” or “111.” (b) Decoding pattern “001” or “110.” (c) Decoding pattern “011” or “100.” (d) Decoding pattern “010” or “101.”

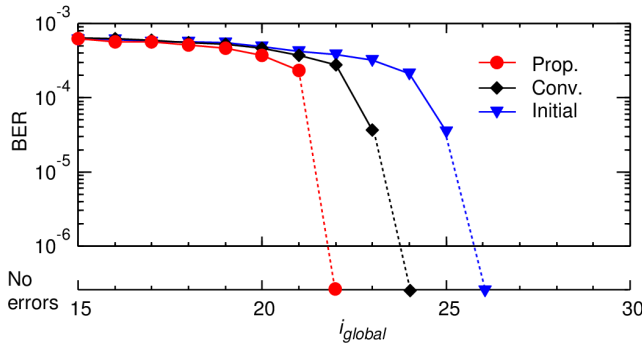


Fig. 11. BER performance for i_{global} . The simulation parameters i_{sp} and SNR_S are set to 20 and 21.0 dB.

proposed fitness function gives a weight of about 1.4 except for “001” or “110,” compared to the conventional one, which contributes to LLR emphasis activations. Figs. 9 and 10 show the selection accuracy and the probability that LLRs are emphasized based on the decoding patterns, respectively. From Fig. 9, we can see that the proposed fitness function gives a slightly lower selection accuracy compared to the conventional one, except for “010” or “101,” but it maintains a high selection accuracy for all decoding patterns. Also, from Fig. 10, the proposed fitness function has a higher probability that LLRs are emphasized for almost all decoding patterns compared to the conventional one. Fig. 11 shows the BER performance for i_{global} . The simulation parameters, i_{sp} and SNR_S , are set to 20 and 21.0 dB, respectively. The blue inverted triangle shows the case where LLRs are emphasized using the initial parameters of the GA. From Fig. 11, the proposed fitness function achieves error-free performance at $i_{\text{global}} = 22$, which is fewer i_{global} compared to otherwise. This result contributes to reducing computational complexity and decoding delay.

From the above, it can be seen that when using GA, a fitness function to improve the selection accuracy and increase the probability that LLRs are emphasized is effective for improving SP decoding performance.

V. CONCLUSION

In this study, to explore more optimal emphasis parameters, we proposed a fitness function to improve the selection accuracy and increase the probability that LLRs are emphasized.

Then, we aimed to improve the performance of SP decoding based on the GA results.

As a result, the proposed fitness function succeeded in increasing the probability that LLRs are emphasized while maintaining a high selection accuracy compared to the conventional one. Furthermore, at $i_{sp} = 20$ and $\text{SNR}_S = 21.0$ dB, the proposed fitness function achieves error-free performance at $i_{\text{global}} = 22$, which is fewer i_{global} compared to the conventional one. Therefore, we clarified that in the GA, setting the fitness function to improve the selection accuracy and increase the probability that LLRs are emphasized is useful for improving SP decoding performance. However, we still have work to do to improve SNR_S .

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